

Relevant Frequency Band Selection using Sequential Forward Feature Selection for Motor Imagery Brain Computer Interfaces

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Abstract— In order to provide basic communication abilities to people with motor disability, motor imagery brain computer interface is one of most widely used technique. In this paper, we present a novel algorithm (Composite Filter Bank based stationary CSP) for determining subject as well as task specific discriminative frequency bands for classification of motor imagery tasks. It is noted in the literature that while performing any motor imagery tasks, two major frequency band of EEG spectrum i.e μ (7-12 Hz) as well as β (12-30 Hz) bands are actively involved. Hence, in most of the literature work EEG signals were filtered using a frequency band of 7-30 Hz usually before using CSP transformation. However, it is possible that some of the frequencies may not provide useful features to distinguish motor imagery tasks. In this paper, we propose a novel approach to select a subset of relevant frequency bands using sequential forward feature selection method from a composite filter bank which consists of Prior-known EEG frequency bands and a set of variable size overlapping frequency bands to improve the performance of motor imagery tasks classification. Experimental results of the proposed work on publicly available datasets validate the effectiveness of the proposed method. Friedman statistical test conducted further shows that the proposed approach significantly outperforms the existing methods.

Keywords— *Common spatial pattern, Motor imagery, Brain computer interface.*

I. INTRODUCTION

Brain computer interface also known as BCI allow individuals having neurological diseases to establish communication and restore motor activity lost due to neurological disorders by converting the EEG signals originating from the brain into device commands [17]. The most used EEG signals that are generated by imagination or actual execution of moving a leg or hand are known as sensorimotor rhythms (μ and β rhythms) or motor imagery signals which originate in primary sensorimotor area [1]. These motor imagery signals are used in Motor imagery BCIs to transform EEG signals into device commands. One of the most widely used and computationally simple data-driven spatial filtering technique in motor imagery BCI is Common Spatial pattern (CSP) method [1, 2]. This method assigns weights to the electrodes placed on brain scalp according to their importance in reflecting the specific activation of cortical areas during motor imagery. Further, it reduces the dimension of the EEG data.

The non-stationary characteristic of EEG signals generating from brain and the artifacts present in raw EEG data proved to be a major drawback while using CSP method which may render noise driven features. Since the number of task-relevant EEG trials is smaller than the number of EEG electrodes, CSP is affected from small size problem [3] as the CSP rendered covariance matrix is of the order of square of the number of electrodes placed on scalp. In CSP analysis, if the number of EEG trials is relatively small, the presence of even a single noisy trial can provide noise driven CSP filter. Under such circumstances of noisy trials, the problem of overfitting occurs [4, 5] which in turn reduces performance of CSP. Another variant of Common Spatial Pattern method that can reduce the effects of the non-stationarity of EEG data and artifacts present is suggested in literature called Stationary CSP (SCSP) [5].

Some research works [6–8] in literature have suggested approaches which use predefined fixed bandwidth filter bank and non-overlapping frequency subband filters to determine CSP spatial filters. Towards this direction, Subband CSP (SBCSP) proposed in the research work of Novi et al. (2007) [6] utilized EEG data filtered from various fixed sized frequency subbands in which features were extracted (CSP and linear discriminant analysis) from each subband to discriminate motor imagery tasks. Recursive feature elimination (RFE) method was used to rank these subband features. In a similar kind of research work, Filter Bank CSP (FBCSP) [7] was proposed that selects optimal spatio-temporal filters by using maximal mutual information criterion to rank features from EEG data filtered from various fixed sized frequency band filters. However, FBCSP as well as SBCSP method use manual selection of subbands in the fixed frequency range (4-40 Hz) having a fixed bandwidth (4 Hz). This reduces the possibility of exploring and utilizing other subject specific relevant subbands of variable size bandwidth that can exist in a given frequency range leading to poor performance. Further in this direction, Combined Variable Size Common Spatial Patterns (CVSCSP) method [8] was proposed that determines the subject-specific relevant frequency bands. In this work, a set of all possible different size frequency subbands were generated for a given bandwidth. However, this method utilized CSP method to determine features from the generated variable sized bands. Thus, CVSCSP cannot mitigate the effects of non-stationarity of the EEG signal and artifacts present. Further, this method does not utilize the existing Prior-known EEG frequency bands that may be relevant for motor imagery classification.

To tackle these drawbacks, in this paper, we proposed to utilize a combination of well-known EEG frequency bands and variable sized overlapping subbands to form a Composite Filter Bank. To obtain features that are more robust to artifacts and non-stationary nature of EEG, SCSP is applied to data filtered from each of the frequency subbands of the Composite Filter Bank. This allows the proposed work to utilize the frequency and spatial content of the EEG data at the same time. In order to find a relevant subset of frequency bands having highly distinguishing features for classification of tasks related to motor imagery, sequential forward feature selection algorithm [9, 10] is utilized in the proposed method. Friedman statistical test [11] is also conducted to investigate statistically significant difference between the proposed CFBCSP method and the state of the art methods i.e. CSP, SCSP, CVSCSP and FBCSP.

The main contributions of the present article include:

- As per author's best knowledge, this is the first method that combines Prior-known EEG Filter Bank and overlapping variable size frequency bands for motor imagery classification.
- We used SFFS method to determine a reduced and relevant set of frequency bands for classification of motor imagery activities.

The remaining portion of this paper is organized as given: The proposed model Composite Filter Bank based SCSP is described in section 2. Section 3 illustrates the results and discussion and section 4 concludes the paper with key contributions of the proposed CFBCSP method.

II. PROPOSED METHOD: COMPOSITE FILTER BANK CSP

The proposed method, Composite Filter Bank based SCSP (CFBCSP) method, involves four phases as shown in Fig. 1. We generate Composite Filter Bank of frequency subbands to filter and pre-process the raw data in the first phase of the proposed approach. To obtain features that are more robust to artifacts and non-stationary nature of EEG, SCSP is utilised to extract features from the bandpass filtered EEG data in the second phase. In the third phase, the sequential forward feature selection based algorithm is opted to determine the most relevant subset of frequency bands. In final phase, SVM classifier is used to build a decision model utilizing the ranked features. Each of these phases is described in detail as given below:

A. Preprocessing

In this phase, first, we formulate different frequency Filter Banks. In the second step, the raw EEG data is bandpass filtered using Butterworth filter from each band of the frequency Filter Banks. In order to obtain the most relevant subset of frequency bands, we combine both the Prior-known EEG Filter Bank and variable size Filter Bank into a single Filter Bank, to form a Composite Filter Bank. Table I, II and III show different sized filter banks (Prior-known EEG Filter Bank [12, 13], Variable Size Filter Bank [8, 14] and Composite Filter Bank) respectively. The Composite Filter Bank utilizes 3 frequency bands [1-4 Hz, 4-7 Hz, and 32-40 Hz] from the Prior-known frequency bands. The other two bands [7-12 Hz and 30-40 Hz] of the fixed sized EEG frequency bands are already present in the variable size bands. Hence the number of subbands in the Composite Filter Bank is 18.

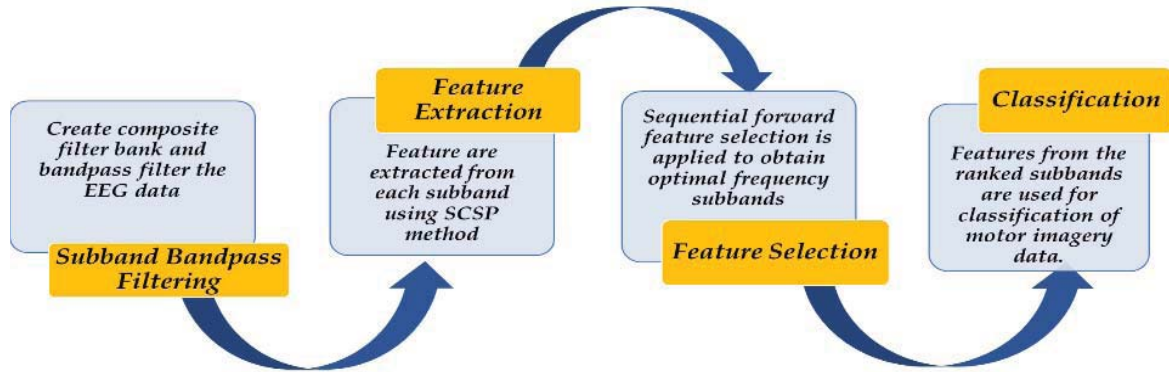


Fig. 1. Steps of the proposed CFBCSP method

TABLE I. PRIOR-KNOWN EEG FILTER BANK

Name	Frequency Band in Hz
<i>delta</i> Band	1-4
<i>theta</i> Band	4-7
<i>alpha</i> Band	7-12
<i>beta</i> Band	12-30
<i>gamma</i> Band	30-40

TABLE II. VARIABLE SIZE FILTER BANK

Name	Frequency Band in Hz
Band 1	7-12
Band 2	7-17
Band 3	7-22
Band 4	7-27
Band 5	7-32
Band 6	12-17
Band 7	12-22
Band 8	12-27
Band 9	12-32
Band 10	17-22
Band 11	17-27
Band 12	17-32
Band 13	22-27
Band 14	22-32
Band 15	27-32

B. Stationary CSP (SCSP)

In CSP algorithm, the covariance matrices of each class EEG data is utilized for simultaneous diagonalization to obtain CSP spatial filters. In CSP, Rayleigh criterion maximization function is optimised to obtain the solution of the simultaneous diagonalization of the covariance matrices:

$$R(\mathbf{W}) = \frac{\mathbf{W}^T \Sigma_1 \mathbf{W}}{\mathbf{W}^T (\Sigma_1 + \Sigma_2) \mathbf{W}} \quad (1)$$

where $\mathbf{W}$ is a spatial filter matrix and Σ_1 and Σ_2 are class 1 and class 2 normalized average covariance matrices respectively. In SCSP, a measure of stationarity is utilized to mitigate the effect of non-stationarity in EEG data, which is computed as the summation of absolute differences among the projected variance of each trial and average of projected variance of all trials. Summation of the difference between the covariance matrix of each trial i and normalized average covariance matrix Σ_1 or Σ_2 for each class c is given by:

$$\Delta_c^{(i)} = s(\Sigma_c^i - \Sigma_c) \quad (2)$$

here s is used as an operator than converts a symmetric matrix positive definite. Average difference matrix for each class $c \in 1, 2$, is given as:

$$\bar{\Delta}_c = \frac{1}{N_c} \sum_{i=1}^{N_c} \Delta_c^i \quad (3)$$

The modified form of the Rayleigh Criterion can be defined as:

$$R(\mathbf{W}) = \frac{\mathbf{W}^T \Sigma_1 \mathbf{W}}{\mathbf{W}^T (\Sigma_1 + \Sigma_2) \mathbf{W} + \mu P(\mathbf{W})} \quad (4)$$

where μ is a constant obtained using cross validation and $P(\mathbf{W}) = \mathbf{W}^T (\bar{\Delta}_1 + \bar{\Delta}_2) \mathbf{W}$ is the penalty term. Using this projection matrix $\mathbf{W}$, the EEG recordings $\mathbf{X}_1^i$ are decomposed into:

$$\mathbf{Z}_i = \mathbf{W} \mathbf{X}_1^i \quad (5)$$

Each column of $\mathbf{W}^{-1}$ matrix is a Common Spatial Patterns whereas each row of $\mathbf{W}$ matrix is a spatial filter. CSP features

TABLE III. COMPOSITE FILTER BANK

Name	Frequency Band in Hz
Band 1	1-4
Band 2	4-7
Band 3	7-12
Band 4	7-17
Band 5	7-22
Band 6	7-27
Band 7	7-32
Band 8	12-17
Band 9	12-22
Band 10	12-27
Band 11	12-32
Band 12	17-22
Band 13	17-27
Band 14	17-32
Band 15	22-27
Band 16	22-32
Band 17	27-32
Band 18	32-40

can be obtained by calculating log variance of the first and last r rows of $\mathbf{Z}_i$ as follows:

$$\mathbf{f}_p = \log \left(\frac{\text{var}(\mathbf{z}_p)}{\sum_{p=1}^{2r} \text{var}(\mathbf{z}_p)} \right) \quad \text{where } \mathbf{p} = 1, 2, \dots, 2r \quad (6)$$

C. Feature Selection

We utilized a wrapper method using sequential forward feature selection search strategy (SFFS) in this work to determine an optimal number of subject-specific frequency subbands for motor imagery classification. In the SFFS method, features are added sequentially to a subset one after the other on the basis of whether adding a feature improves the overall performance of the subset. The process is repeated until the performance starts degrading. We used classification accuracy of the decision system using SVM classifier as a performance measure in this work.

The whole framework for sequential forward feature selection method for obtaining a relevant and non-redundant subset of features is given below:

Sequential Forward Feature Selection (SFFS)

Input: Set of d frequency band based features $\mathbf{F} = \{f_1, f_2 \dots f_d\}$,

Output: Subset of frequency band features $\mathbf{s}$

Step 1: Initialization

Start with an empty subset of relevant features: $\mathbf{S} = \emptyset$

Initiate performance metric as: $\text{SVM}_{\text{acc}}(\mathbf{S}) = 0$

Step 2: Feature subset generation and evaluation:

For each feature f_k in $\mathbf{F}$, add generated candidate feature subset by adding one feature at a time to the existing best subset and select single feature f_k such that

$$f_k = \max_{f_i} \text{SVM}_{\text{acc}}(\mathbf{S} \cup f_i)$$

$$\mathbf{S} = \mathbf{S} \cup f_k$$

$$\mathbf{F} = \mathbf{F} - f_k$$

Step 3: Repeat step 2 until:

$$\text{SVM}_{\text{acc}}(\mathbf{S}(k)) < \text{SVM}_{\text{acc}}(\mathbf{S}(k-1))$$

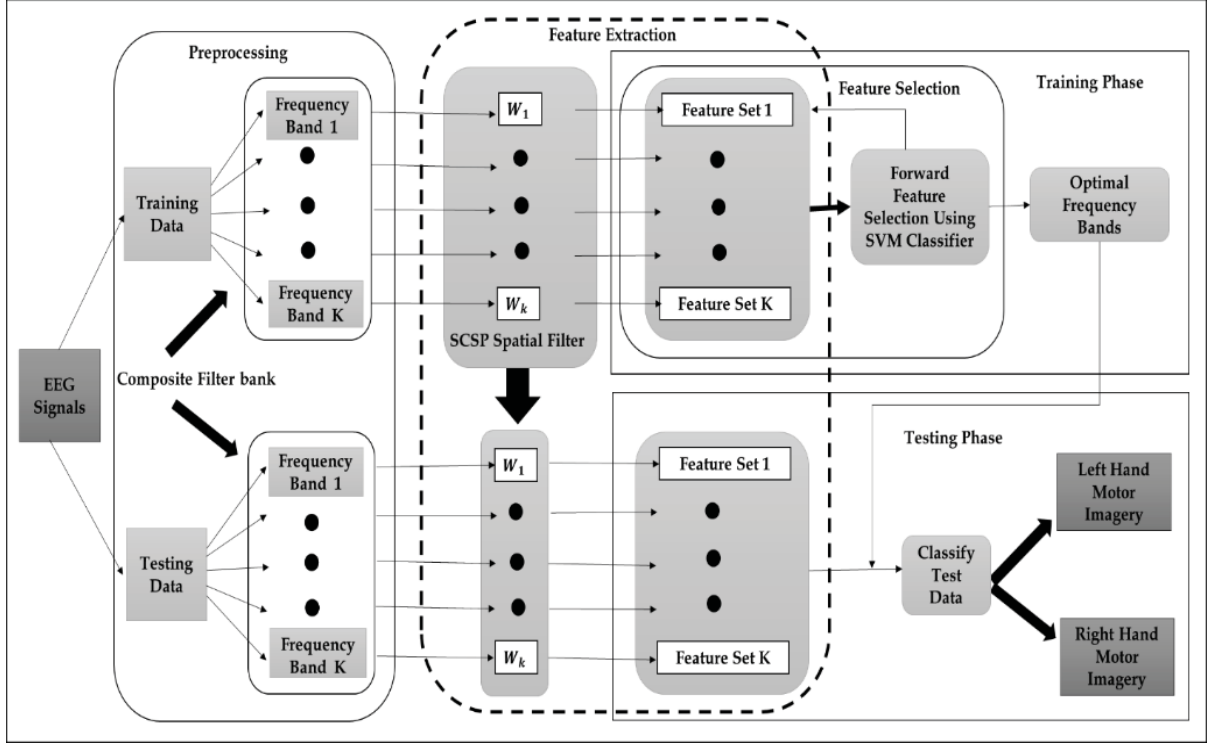


Fig. 2. The block diagram of the proposed CFBCSP method

III. EXPERIMENTAL RESULTS

The steps of the proposed CFBCSP method is shown in Fig. 2. To investigate the effectiveness of the proposed framework, experiments are carried out on two publicly available Datasets:

- (i) Dataset 4a from BCI Competition III (Dataset 1) [15]
- (ii) Dataset 1a from BCI competition IV (Dataset 2) [16]

In our experiments, EEG data of each trial belonging to the time slot [0.5 to 2.5 sec] after the commencement of stimulus is utilised which yields in an EEG signal matrix having 118 channels as rows and 200 time points as columns (118×200) per trial for Dataset 1. Similarly, a data matrix of (59×200) per trial for Dataset 2 was obtained. To explore the relevance of frequency bands for motor imagery classification, we have performed experiments on Prior-known EEG Filter Bank and Composite Filter Banks separately.

Number of frequency bands in Prior-known EEG Filter Bank, variable size Filter Bank, and Composite Filter Bank is 5 and 18 respectively. These variable size frequency bands were obtained using the method given in [8] from a frequency range of [7-30 Hz]. Firstly, the raw data is bandpass filtered using these three subbands Filter Banks in the pre-processing stage. In the feature extraction step, SCSP is utilised to extract features. SCSP penalty parameter $\mu = 0.1$ was used for all the experiments on both Datasets. The count of spatial patterns was fixed to $r = 1$ and thus, we attained two features from each trial. Sequential forward feature selection algorithm is utilized for identifying relevant and optimal subset of

frequency bands. For optimizing the performance of SVM based learning model, we employed grid search to obtain the Gaussian kernel parameter σ and SVM regularization parameter C which varied from 0.01 to 100 and 1 to 1000 respectively. $C=100$ and $\sigma=10$ were obtained optimal values based on experimentation.

The classification accuracy is adopted as performance measure for the proposed CFBCSP method and compared it with existing state of the art algorithms (CSP, SCSP, FBCSP, and CVSCSP). The classification performance of the proposed CFBCSP method is further compared with its own variants with different size Filter Banks. For each subject, an average classification accuracy of 10 runs of 10-fold cross-validation is used to report the classification accuracy metric used in this article as.

Table IV shows the classification accuracy obtained using various methods for different subjects of Dataset 1. Bar charts shown in Fig. 3 depicts an improvement in percentage classification accuracy of the proposed CFBCSP method in comparison to other methods for all subjects of Dataset 1.

TABLE IV. CLASSIFICATION ACCURACY OBTAINED FROM DIFFERENT METHODS (DATASET 1)

Subject	Without Feature Selection		Filter Approach		Wrapper Approach	
	CSP	SCSP	FBCSP	CVSCSP	Prior-known	CFBCSP

					EEG Filter Bank	
aa	75.37	80.45	80.67	80.26	80.76	84.60
al	97.73	94.38	97.05	97.51	94.69	98.49
av	69.14	69.82	69.07	72.56	74.40	74.40
aw	82.27	82.43	83.70	82.30	81.60	85.37
ay	82.17	89.33	63.33	85.50	85.33	90.67
Mean	81.34	83.28	78.76	83.63	83.36	86.70

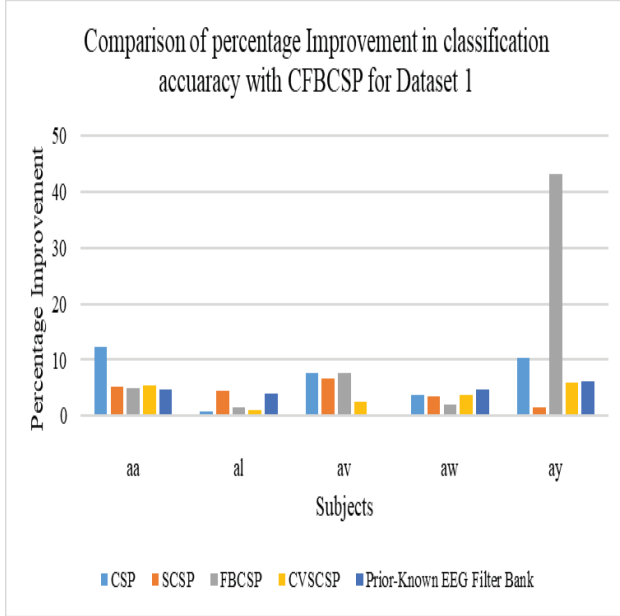


Fig. 3. Comparison of improvement in classification accuracy with CFBCSP (Dataset 1)

Following observations can be made from Table IV and Fig. 3 for Dataset 1:

- The proposed CFBCSP method achieved maximum classification accuracy.
- CFBCSP outperforms the standard CSP, SCSP, FBCSP, CVSCSP and Prior-known EEG Filter Bank for all subjects in terms of classification accuracy.
- There is an overall increase of **4.11%, 6.60%, 10.08%, 3.68% and 4.01%** in average classification accuracy of the proposed CFBCSP method in comparison to SCSP, CSP, FBCSP, CVSCSP and Prior-known EEG Filter Bank approach respectively.

TABLE V. CLASSIFICATION ACCURACY OBTAINED FROM DIFFERENT METHODS (DATASET 2)

Subject	Without Feature Selection	Filter Approach	Wrapper Approach
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	CSP	SCSP	FBCSP	CVSCSP	Prior-known EEG Filter Bank	CFBCSP
ds1a	73.10	81.75	72.30	69.70	65.65	69.60
ds1b	65.40	59.95	54.70	58.85	64.65	65.20
ds1c	70.50	75.10	76.00	80.60	77.55	79.45
ds1d	76.80	89.20	91.70	90.85	95.36	96.75
ds1e	83.30	90.35	85.75	90.10	93.55	94.95
ds1f	82.80	86.90	81.00	82.05	86.85	87.05
ds1g	79.00	91.45	91.45	92.50	92.55	93.65
Mean	75.84	82.10	78.99	80.66	82.31	83.81

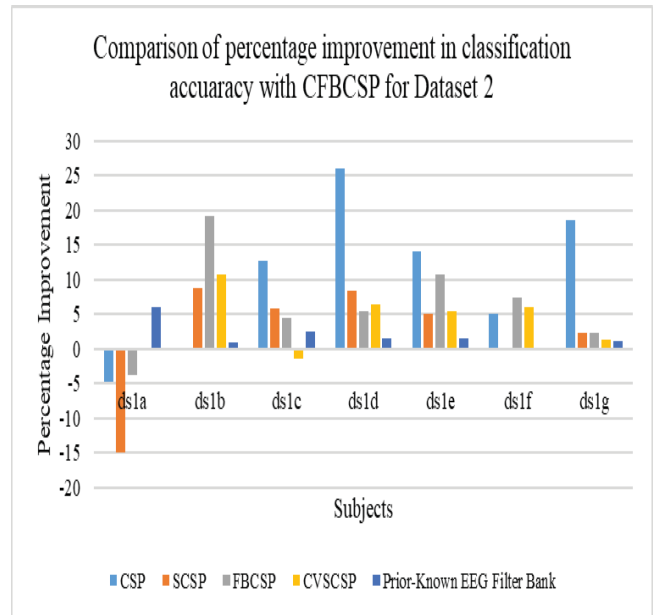


Fig. 4. Comparison of improvement in classification accuracy with CFBCSP (Dataset 2)

Table V shows the classification accuracy obtained using various methods for different subjects of Dataset 2. Bar charts shown in Fig. 4 depicts an improvement in percentage classification accuracy of The proposed CFBCSP method in comparison to other methods for all subjects of Dataset 2.

Following observation can also be derived for Dataset 2 from Table V and Fig. 4:

- The proposed CFBCSP method achieved maximum classification accuracy.
- The classification accuracy with CFBCSP method outperforms the standard CSP for all the subjects, (with the exception of subject *ds1a* and *ds1b*)

- The classification accuracy with CFBCSP outperforms the SCSP and FBCSP for all the subjects (with the exception of subject *ds1a*).
- The classification accuracy with CFBCSP outperforms CVSCSP for all the subjects, (with the exception of subject *ds1a* and *ds1c*).
- The classification accuracy with CFBCSP outperforms Prior-known EEG Filter Bank approach for all the subjects.
- For all subjects, there is an overall increase of **2.07%, 10.50%, 6.10%, 3.89% and 1.82%** in average classification accuracy of The proposed CFBCSP method in comparison to SCSP, CSP, FBCSP, CVSCSP, Prior-known EEG Filter Bank approach.

The following can be concluded from the above observations:

- There is an overall increase of **8.83%, 3.07%, 8.77%, 3.92% and 2.77%** in average classification accuracy of The proposed CFBCSP method in comparison to CSP, SCSP, FBCSP, CVSCSP and Prior-known EEG Filter Bank approach.
- Classification accuracy improves with the employment of feature selection.

- Wrapper feature selection approach (SFFS) performs better than filter feature selection approach.

Boxplot in Fig. 5 shows variations in average classification accuracy of 12 subjects of both Datasets of all the methods. It can be observed from Fig. 5 that the median accuracy is very similar for Composite Filter Bank method (CFBCSP) and SCSP method. However, upper quartile of CSP and SCSP show a decrease in performance as compared to Fixed sized Filter Bank, and CFBCSP.

Fig. 6 and Fig. 7 show relevant subset of frequency bands obtained by the proposed CFBCSP method for motor imagery classification for each subject of Dataset 1 and for each subject of Dataset 2 respectively.

We can finally conclude From Fig. 6 and Fig. 7 that both Prior-known frequency bands and Variable Size bands are relevant for both Datasets for motor imagery classification. The Composite Filter Bank allows us to investigate the relevance of both Prior-known EEG Filter Bank and Variable Size Filter Bank to motor imagery classification. It is also observed that the frequency filters are subject dependent as well as task dependent.

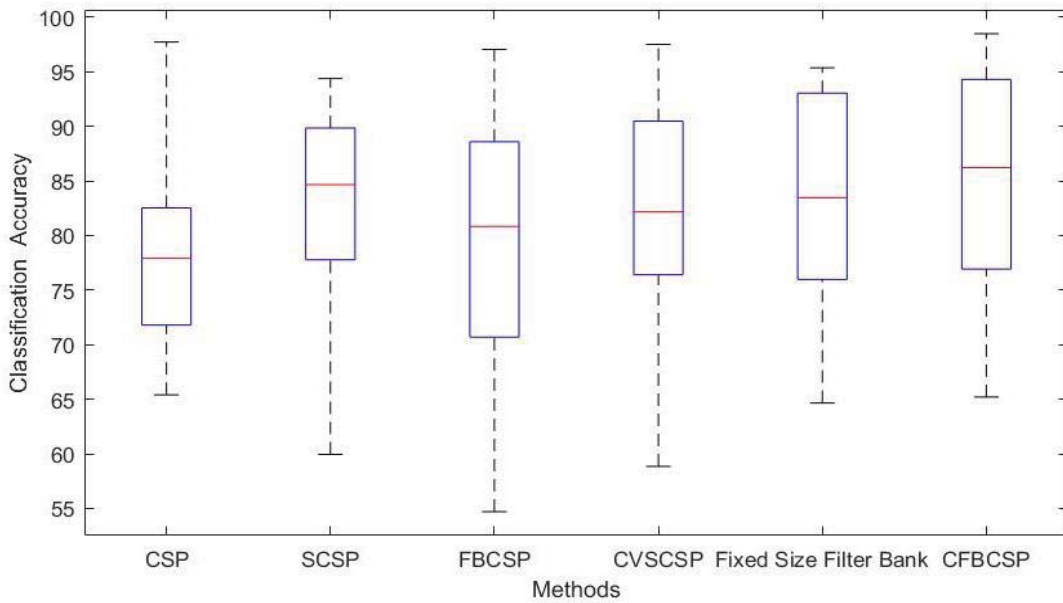


Fig. 5. Boxplot of average classification accuracy over all subjects of both Datasets from different methods

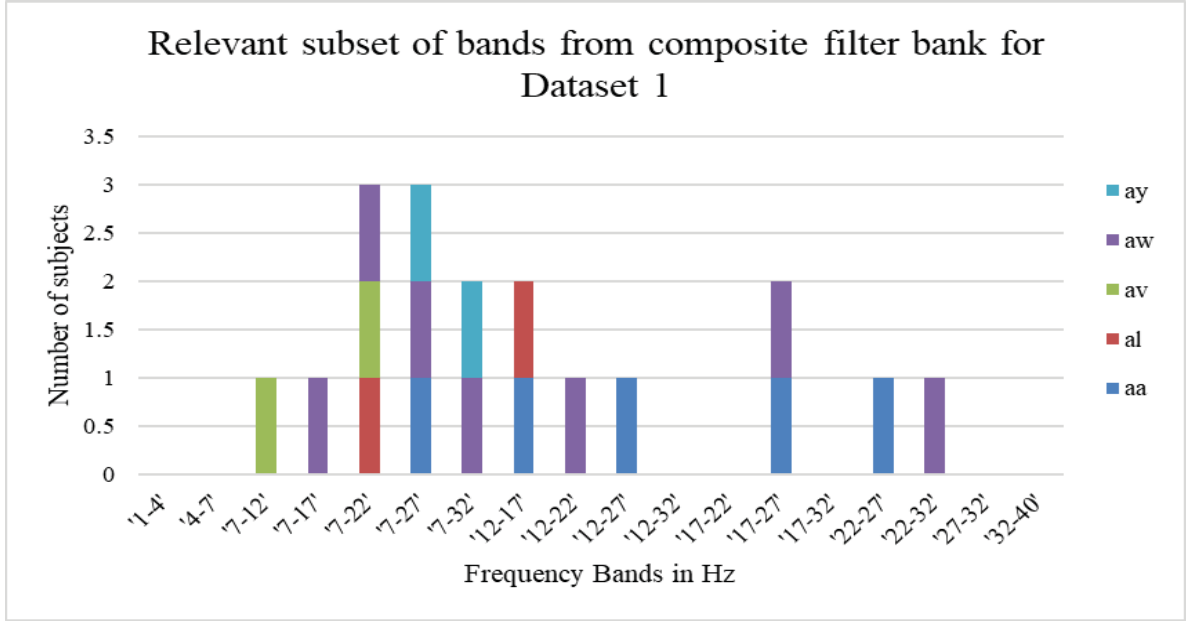


Fig. 6. Relevant subset of bands from Composite Filter Bank for Dataset 1

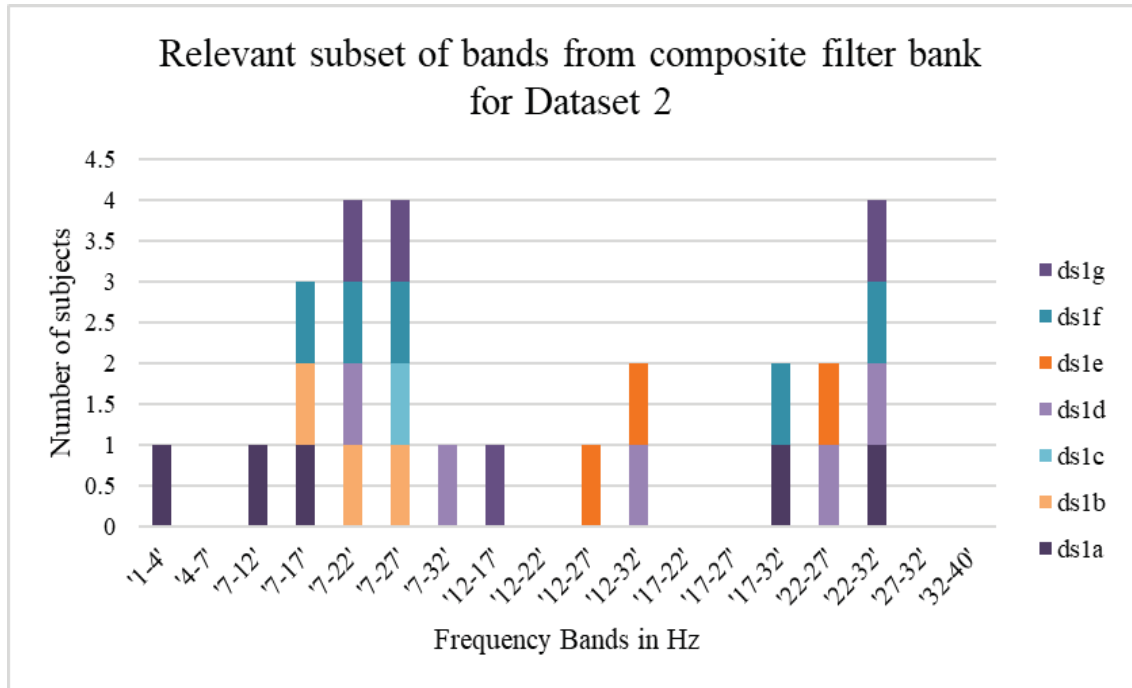


Fig. 7. Relevant subset of bands from Composite Filter Bank for Dataset 2

A. Statistical test

In this work, the statistical significant difference among all the seven methods is specified by conducting Friedman statistical tests[11]. The Friedman ranking for all the methods over all Datasets is given in Table VI. It can be observed from Table VI that the proposed CFBCSP method achieved the least rank value of 1.95.

TABLE VI. FRIEDMAN RANKING OF METHODS

Method	Friedman Ranking
CFBCSP	1.95
<i>Prior-known EEG Filter Bank</i>	4.25
<i>CVSCSP</i>	4.50
<i>SCSP</i>	4.54
<i>FBCSP</i>	5.29
<i>CSP</i>	5.33

The p -value computed from Iman and Davenport statistic is $3.63\text{E-}5$ suggests that there exists significant statistical difference among the classification accuracy of all the seven methods compared in this experiment.

IV. CONCLUSION

We proposed a novel method for motor imagery tasks classification using a Composite Filter Bank approach in this research work. The proposed model consists of four phases. In the first phase, data is pre-processed using bandpass filtering from different sized Filter Banks. In the second phase, features are extracted using SCSP from each of the frequency bands of different Filter Banks. The decision model is learned on an optimal subset of features determined using sequential forward feature selection method.

We evaluated the proposed model on two publicly available Datasets and compared with existing models CSP, SCSP, FBCSP and CVSCSP. The proposed model outperforms existing methods with a classification accuracy of 86.7% (Dataset 1) and 83.8 % (Dataset 2) . To illustrate the statistical significance of the proposed CFBCSP method, Friedman statistical method is used.

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